**1.** Ligth from a helium-neon laser ($\lambda = 633$ nm) passes through a narrow slit and is seen on a screen 2.0 m behind the slit. The first minimum in the diffraction pattern is 1.2 cm from the central maximum. How wide is the slit?

**2.** Light passes through a 0.12-mm-wide slit and forms a diffraction pattern on a screen 1.00 m behind the slit. The width of the central maximum is 0.85 cm. What is the wavelength of light?

**3.** You need to use your cell phone, which broadcasts an 800 MHz signal, but your behind two massive, radio-wave-absorbing buildings that have only a 15 m space between them. What is the angular width, in degrees, of the electromagnetic wave after it emerges from the building. Assume only the central maximum.

**4.** a 500 line/mm diffraction grating is illuminated by light of wavelength 510 nm. How many bright fringes are seen on a 2.0-m-wide screen located 2.0 meters behind the diffraction grating?

**5.** White light(400 -700 nm) incident on a 600 line/mm diffraction grating produces rainbows of diffracted light. What is the width of the first order rainbow seen on a screen 2.0 m behind the grating?

**6.** Use GeoGebra to graph the intensity pattern for single-slit diffraction. The intensity is given by $I(y) = I_0 \operatorname{sinc}^2(\beta)$ where $\beta = \frac{\pi a y}{\lambda L}$ and $y$ is the position on the screen.

- **A.** Create sliders for wavelength $\lambda$ (400–700 nm), slit width $a$ (1–20 $\mu$m), and screen distance $L$ (20–100 cm).
- **B.** Plot the intensity pattern. In GeoGebra, use `sinc(x) = sin(x)/x` (note: handle $x = 0$ separately or GeoGebra may do it automatically).
- **C.** Set $\lambda = 620$ nm, $a = 4$ $\mu$m, $L = 40$ cm. Sketch or screenshot your result.
- **D.** From your graph, estimate the positions of the first and second minima. Verify these match the formula $a \sin\theta = m\lambda$.